

Machine Learning Fundamentals

Machine learning is a subset of artificial intelligence that focuses on building systems capable of learning from data rather than being explicitly programmed. There are three broad categories of machine learning: supervised learning, unsupervised learning, and reinforcement learning. Supervised learning involves training a model on labeled examples, where each input is paired with a correct output, allowing the model to learn a mapping between the two. Common supervised learning tasks include classification, where the goal is to assign inputs to discrete categories, and regression, where the goal is to predict continuous numerical values. Unsupervised learning, on the other hand, works with unlabeled data and aims to discover hidden structure, such as grouping similar data points together through clustering algorithms. Reinforcement learning takes a different approach entirely, training an agent to make decisions by rewarding desirable actions and penalizing undesirable ones within an environment. Popular algorithms in machine learning include decision trees, support vector machines, logistic regression, and increasingly, deep neural networks. The choice of algorithm often depends on the size and nature of the dataset, the complexity of the problem, and the computational resources available. A critical part of any machine learning pipeline is evaluation, where metrics such as accuracy, precision, recall, and F1 score are used to measure how well a model performs on unseen data. Overfitting, where a model performs well on training data but poorly on new data, remains one of the most common challenges practitioners must guard against.